COMPUTER PROGRAMING LAB REPORT # 1

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DEPARTMENT	ELECTRICAL ENG (power)
SECTION	“A”
SEMESTER	1st
SUBJECT	COMPUTER PROGRAMING

1. Objective :

Clearly state what you intended to learn or achieve in the lab session.

- *Example:* "To understand the basic structure of a C++ program and perform simple arithmetic operations".

3.Discussion of Fundamentals (Theory) :

Briefly explain the key concepts used in the lab.

- **Headers:** Why #include <iostream> is used (for input/output).
- **Namespace:** The purpose of using namespace std;.
- **Main Function:** Explanation of int main() as the program's entry point.
- **Variables/Data Types:** Brief overview of int, float, or char used.

4. *Problem Statement (Task Description)*

Reiterate the specific programming tasks you were assigned word-for-word from the lab manual.

- Example:
- "Write a program that takes two numbers and displays their sum".

5. *Implementation (Code)*

Paste your source code here. Use a monospaced font (like **Consolas**) to keep it readable.

- ***Requirement:***
- Include comments in your code to explain specific logic.

6. **Experimental Results (Output)**

Provide evidence that your program worked correctly.

- ***Screenshots***: Include a clear image of the console window showing the output.
- ***Test Cases***: Show how the program handles different inputs (e.g., positive numbers, zero, or invalid characters).

7. *Analysis & Conclusion*

- ***Discussion***:
- Explain how the program works, your choice of variable types, and how loops or conditions executed.
- ***Difficulties:***
- Mention any bugs (syntax or logic errors) you encountered and how you fixed them.
- ***Lessons Learned:***
- Summarize the skills acquired, such as mastering cout/cin or basic arithmetic.

EXAMPLE # 1

```
#include <iostream> // Header for input/output
using namespace std;

int main() {
    // Prints the string to the console
    cout << "Hello, World!" << endl;

    return 0;
}
```

EXAMPLE # 2

```cpp
#include <iostream>
using namespace std;

int main() {
    int num1, num2, sum;

    cout << "Enter first integer: ";
    cin >> num1;
    cout << "Enter second integer: ";
    cin >> num2;

    sum = num1 + num2; // Addition logic

    cout << "The sum of " << num1 << " and " << num2 << " is: " << sum << endl;

    return 0;
}
```

EXAMPLE # 3

```cpp
#include <iostream>
using namespace std;

int main() {
    float radius, area;
    const float PI = 3.14159; // Constant value for Pi

    cout << "Enter the radius of the circle: ";
    cin >> radius;

    area = PI * radius * radius; // Formula: PI * r^2

    cout << "The area of the circle is: " << area << endl;

    return 0;
}
```